

Dr. NKODIA Hardy Medry Dieu-Veil

Geologist — Tectonics, geodynamics and structural geology

Congolese nationality • Brazzaville, Republic of Congo

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PROFILE

Lecturer–researcher in structural geology and tectonics, specialising in palaeostress analysis, fault reactivation and intraplate seismic hazard in Central Africa. Field-based approach combined with remote sensing and stress modelling (*slip tendency*).

RESEARCH INTERESTS

Fault/fracture reservoir architecture (Inkisi, Pool region); geomorphology and erosion of the Congo Basin; mineralisations, rock deformation and aquifers; seismic hazard and fault kinematics on the Central African rifted margin.

EDUCATION

2019–2024	PhD in Geosciences — Marien Ngouabi University (Congo) & Royal Museum for Central Africa (Belgium). Tectonics, geodynamics and structural geology. Highest distinction with the jury's congratulations.
2015–2017	MSc in Geosciences — Marien Ngouabi University (Congo). Distinction <i>Bien</i> ; top of class.
2012–2015	BSc in Geosciences — Marien Ngouabi University (Congo). Distinction <i>Assez bien</i> ; 2nd of class.
2011	Baccalauréat (series D) — Lycée Chaminade, Brazzaville. Distinction <i>Bien</i> ; 1st of school.

CERTIFICATIONS

2023–2024	Professional Certificate — Project Management — Google (online).
2023	Writing and publishing a scientific paper (project-based) — École Polytechnique, Paris (online).
2022	Wetland mapping — Remote Sensing & Forest Ecology Lab, ENS, Congo.
2019	Introduction to radar remote sensing — IGNFI / Marien Ngouabi University / AFD.

ACADEMIC AND PROFESSIONAL APPOINTMENTS

Assistant (Lecturer–Researcher)

2025 – present

Marien Ngouabi University, Faculty of Science and Technology, Dept. of Geology (Congo)

- Tenure-track position, prior to promotion to Assistant Professor.
- Structural Geology lecture course (L2); Petrology–Mineralogy practicals.
- Supervision of MSc theses; design of project-based teaching.

Postdoctoral researcher

2025 – 2026

Pukyong National University, Busan (South Korea) — on research leave from Marien Ngouabi University

- Contemporary stress field and seismogenic structures of the Korean Peninsula: focal-mechanism inversion and slip-tendency analysis.

Research visit

Aug – Sep 2026

Royal Museum for Central Africa (RMCA), Tervuren (Belgium)

- West Congo Belt and karst systems.

Adjunct Lecturer–Researcher

2018 – 2024

Marien Ngouabi University, Faculty of Science and Technology (Congo)

- Taught structural geology; co-organised an international geology conference.
- Author / co-author of several peer-reviewed international journal articles.

Postdoctoral researcher (research visit)

Feb–Mar 2025

Royal Museum for Central Africa (RMCA), Tervuren (Belgium)

- Structural analysis and palaeostress reconstruction of the Central African rifted margin.

Research assistant

2019 – 2021

Royal Museum for Central Africa (RMCA), Tervuren (Belgium)

- Co-organised the 1st international conference “Geology and natural resources in Central Africa: societal impact and sustainable development in Congo”.
- Supported the organisation of a Tectonic Studies Group meeting (London).
- Managed a €15,000 research budget over 3 years; international collaborations.

Instructor

May 2019 – Jan 2021

African School of Development (EAD), Brazzaville (Congo)

- Introduced and taught the structural geology course in the curriculum.

Founder & President

2019 – present

Kongo Science, Brazzaville (Congo)

- Congo’s first digital scientific library; capacity-building for African researchers.
- Online and in-person courses; workshop with Prof. Francine Ntouni gathering nearly 500 students.

GRANTS AND PROJECTS

- **AMORCE-GEO (2025–2027)** — cooperation project funded by ARES (Belgium). Role: co-organiser and trainer.
- **RMCA research budget** — €15,000 over 3 years (2019–2021).

HONOURS AND DISTINCTIONS

- PhD awarded with highest distinction and the jury’s congratulations (2024).
- Top of class — MSc in Geosciences (2017).
- 2nd of class — BSc in Geosciences (2015).
- 1st of school — Baccalauréat (2011).

PUBLICATIONS — PEER-REVIEWED ARTICLES

1. **Nkodia, H. M. D.-V.**, Hategekimana, F., Naik, S. P., Cheon, Y., Peace, A. L., Bae, S.-Y., Kandregula, R. S., & Kim, Y.-S. (2026). Vertical kinematic partitioning in intraplate fault systems: evidence from the SE Korean Peninsula. *Journal of Structural Geology*, **211**, 105778. doi.org/10.1016/j.jsg.2026.105778
2. **Nkodia, H. M. D.-V.**, Park, K., Naik, S. P., Peace, A. L., & Kim, Y.-S. (2026). Assessing the contemporary stress field and seismogenic structures of the Korean peninsula using focal-mechanism inversion and slip-tendency analysis. *Tectonophysics*, **934**, 231261. doi.org/10.1016/j.tecto.2026.231261
3. Colet, M., Kolawole, F., Ajala, R., Delvaux, D., & **Nkodia, H. M. D.-V.** (2025). Active crustal deformation across a nucleating extensional microplate, D. R. Congo, East Africa. *Tectonics*, **44**, e2025TC008815. doi.org/10.1029/2025TC008815
4. Mavoungou, Y. B., **Nkodia, H. M. D.-V.**, Watha-Ndoudy, N., & Bolarinwa, A. T. (2024). Lineament extraction and paleostress analysis in the Bikélélé iron deposit (Chaillu Massif, Republic of Congo). *Modeling Earth Systems and Environment*, **10**(2), 1993–2009. doi.org/10.1007/s40808-023-01883-3
5. Miyouna, T., Florent, B., **Nkodia, H. M. D.-V.**, & Delvaux, D. (2024). The Cambro-Ordovician Gondwana alluvial megafan in Central Africa: Palaeozoic sandstones of the Inkisi Group. *Journal of African Earth Sciences*, **209**, 105109. doi.org/10.1016/j.jafrearsci.2023.105109
6. **Nkodia, H. M. D.-V.**, Boudzoumou, F., Miyouna, T., Kongota, E., Ganza, G. B., Lahogue, P., & Delvaux, D. (2024). Brittle faulting and tectonic stress history on the western margin of the Congo Basin between Kinshasa and Brazzaville. *Tectonophysics*, **877**, 230282. doi.org/10.1016/j.tecto.2024.230282
7. Loemba, A. P. R., Ferreira, A., Ntsiele, L. J. E. P., Opo, U. F., Bazebizonza Tchiguina, N. C., **Nkodia, H. M. D.-V.**, Klausen, M., & Boudzoumou, F. (2023). Archean crustal generation and Neoproterozoic partial melting in the Ivindo basement, NW Congo craton. *Journal of African Earth Sciences*, **200**, 104852. doi.org/10.1016/j.jafrearsci.2023.104852

8. **Nkodia, H. M. D.-V.**, Miyouna, T., Kolawole, F., Boudzoumou, F., Loemba, A. P. R., Bazebizanza Tchiguina, N. C., & Delvaux, D. (2022). Seismogenic fault reactivation in western Central Africa: insights from regional stress analysis. *Geochemistry, Geophysics, Geosystems*, 23(11), e2022GC010377. doi.org/10.1029/2022GC010377
9. **Nkodia, H. M. D.-V.**, Miyouna, T., Delvaux, D., & Boudzoumou, F. (2020). Flower structures in sandstones of the Palaeozoic Inkisi Group (Brazzaville, Republic of Congo). *South African Journal of Geology*, 123(4), 531–550. doi.org/10.25131/sajg.123.0038
10. Bazebizanza, N. C., Miyouna, T., **Nkodia, H. M. D.-V.**, & Boudzoumou, F. (2020). Detection of neotectonic signatures by morphometric analysis of the Inkisi Group on both banks of the Congo River. *Journal of Geosciences and Geomatics*, 8, 83–93. doi.org/10.12691/jgg-8-2-4
11. Miyouna, T., Bazebizanza, N. C., Florent, E., Kempena, A., **Nkodia, H. M. D.-V.**, & Boudzoumou, F. (2020). Satellite image mapping of lineaments of the Inkisi Group, Republic of Congo. (16), 68–84.
12. Nkodia, A., Bazebizanza, N. C., & **Nkodia, H. M. D.-V.** (2020). Epidemiological features and spatio-temporal dynamics of the COVID-19 pandemic in the Republic of Congo, 2020. (21), 39–45.
13. Miyouna, T., **Nkodia, H. M. D.-V.**, Essouli, O. F., Dabo, M., Boudzoumou, F., & Delvaux, D. (2018). Strike-slip deformation in the Inkisi Formation, Brazzaville, Republic of Congo. *Cogent Geoscience*, 4(1), 1542762. doi.org/10.1080/23312041.2018.1542762

PREPRINTS

- Loemba, R., **Nkodia, H. M. D.-V.**, Opo, U. F., Bazebizanza Tchiguina, N. C., Miyouna, T., & Boudzoumou, F. (2022). A unified structural framework for major shear zones in the Ntem–Chaillu–Ivindo blocks. *SSRN preprint*. doi.org/10.2139/ssrn.4220669

INVITED TALKS

- **Invited seminar** — “Earthquake Hazards of Rifted Margins” online series, Columbia University / Lamont–Doherty Earth Observatory (convener: Dr. F. Kolawole), **6 July 2026**. Talk: “Ocean–continent tectonic stresses and earthquake hazards of the Central African rifted margin”. Recording: YouTube.

CONFERENCE PRESENTATIONS

- **Nkodia, H. M. D.-V.**, Miyouna, T., Boudzoumou, F., & Delvaux, D. (2021). Integration of flower structures in strike-slip fault damage zones classification. EGU General Assembly, Vienna.
- **Nkodia, H. M. D.-V.**, Boudzoumou, F., Miyouna, T., Ibarra-Gnianga, A., & Delvaux, D. (2021). A progressive episode of deformation in the foreland of the West-Congo Belt. EGU General Assembly, Vienna.
- **Nkodia, H. M. D.-V.**, Delvaux, D., Boudzoumou, F., & Miyouna, T. (2021). Paleostress reconstruction of brittle deformation in the foreland of the West-Congo Belt. Geologica Belgica International Meeting, Tervuren (Belgium).
- Delvaux, D., Miyouna, T., Boudzoumou, F., & **Nkodia, H. M. D.-V.** (2022). Combined paleostress reconstruction and slip tendency for a complex strike-slip fault system: the Pool area. EGU General Assembly, Vienna.
- Arfaoui, I., Bazebizanza, N. C., Thijenira, A., Ngala, N., Lutete, J., Samba, P., **Nkodia, H. M. D.-V.**, Boudzoumou, F., Delvaux, D., & Lahogue, P. (2022). New inventoried karst systems in Central Africa and preliminary palaeoclimate proxy analysis.
- Loemba, A. P., **Nkodia, H. M. D.-V.**, Bazebizanza, N. C., Miyouna, T., & Boudzoumou, F. (2022). Tectonic and structural evolution of a major shear zone in the Ntem–Chaillu Block. Tectonic Studies Group Annual Meeting.
- Loemba, A. P., Legrand Juldit Espoir, P., Fiacre, O., Bazebizanza, N. C., **Nkodia, H. M. D.-V.**, & Boudzoumou, F. (2022). Crustal growth of Archean and early Proterozoic granitoids of the Ivindo region. EGU General Assembly, Vienna.
- Delvaux, D., Gandza, G., Kongota, E., **Nkodia, H. M. D.-V.**, Miyouna, T., & Boudzoumou, F. (2019). The “fault of the Pool” along the Congo River is no longer a myth. EGU General Assembly, Vienna.
- **Nkodia, H. M. D.-V.**, Miyouna, T., Boudzoumou, F., Mbilou, U., Kongota, E., & Florent, E. (2018). Structural and tectonic style of the Inkisi Formation in the Brazzaville area. Congo Geology Conference, Kinshasa.

SCIENTIFIC SERVICE AND OUTREACH

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- Founder of the Laboratory of Geodynamics and Natural Resources (LGRN), Marien Ngouabi University — drafting of statutes and internal rules.
 - President of Kongo Science (scientific community, since 2019).
 - Science outreach and media coverage (press: Dépêches de Brazzaville, Le Papyrus).

SKILLS AND LANGUAGES

- Tectonic and structural analysis; palaeostress reconstruction and seismicity analysis (*slip tendency*).
- Remote sensing and GIS mapping; lineament extraction (ALOS-PALSAR DEM, satellite and drone imagery).
- Petrology of sedimentary, igneous and metamorphic rocks; thin-section preparation; field geology.
- Scientific illustration (Adobe Illustrator); Python for publication-quality figures (600 dpi).
- Project management, budgeting and cost analysis; online instructional design.
- Languages: French (native), English (professional working proficiency).

REFERENCES

- **Prof. Florent Boudzoumou** — Marien Ngouabi University.
- **Prof. Averti Ifo** — Marien Ngouabi University.
- **Dr. Damien Delvaux** — Senior Geologist; Honorary Head, Geodynamics and Mineral Resources Unit; Editor-in-Chief, *Journal of African Earth Sciences*; Royal Museum for Central Africa, Tervuren (Belgium).
- *Referees' contact details available on request.*